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AI adoption and cross-border e-commerce in a fragmented and uncertain European Union

Report

1. Introduction and motivation

Digital transformation is reshaping economic activity across the European Union, with artificial intelligence (AI) playing an increasingly important role in business operations. At the same time, the adoption of AI remains uneven across EU member states, contributing to technological fragmentation within the Single Market.

Cross-border e-commerce is a key channel through which digitalisation supports economic integration. By enabling firms to reach customers beyond national borders, online trade can reduce traditional barriers to market access. However, participation in cross-border e-commerce differs widely across countries, reflecting disparities in digital capabilities and firms' readiness to operate online. Understanding the drivers of these differences is therefore of direct relevance for EU economic and digital policy.

The relationship between AI adoption and cross-border e-commerce is particularly relevant in the current economic environment, which is characterised by geopolitical tensions and heightened macroeconomic uncertainty. Exchange rate volatility and external shocks further affect cross-border commercial activity. In this context, digital technologies may support firms' ability to adapt and remain competitive in international online markets.

Against this background, this project examines whether differences in AI adoption across EU countries are associated with variation in cross-border e-commerce outcomes, with a particular focus on business-to-consumer e-commerce. The central research question is: to what extent is AI adoption associated with cross-border e-commerce performance across EU member states in a fragmented and uncertain economic environment?

2. Data sources and methodology

The analysis uses external statistics from Eurostat and the European Central Bank (ECB). Data on artificial intelligence (AI) adoption are drawn from Eurostat surveys on ICT usage in enterprises and measure the share of firms using AI technologies across EU member states. Two indicators are employed: firms using at least one AI technology, which serves as the main measure of AI adoption, and firms using at least two AI technologies, which is used as a robustness check.

E-commerce indicators are also taken from Eurostat and capture the share of enterprises engaged in business-to-consumer (B2C) and business-to-business (B2B) online sales. B2C e-commerce is treated as the main outcome variable due to its direct relevance for cross-border digital trade, while B2B e-commerce is included for comparison. Exchange rate data from the ECB are used to provide macroeconomic context and reflect the uncertainty surrounding cross-border economic activity.

The analysis follows a descriptive and comparative approach. Cross-country differences are examined using maps and scatter plots, while selected time-series evidence illustrates heterogeneous development paths across countries. The focus is on identifying associations between AI adoption and e-commerce outcomes rather than establishing causal relationships.

This approach allows for a consistent comparison across countries while remaining transparent about the descriptive nature of the analysis.

3. Empirical findings

The analysis reveals pronounced cross-country differences in AI adoption across the European Union. Northern and digitally advanced economies display consistently higher levels of AI use among enterprises, while several Central and Eastern European countries lag behind. This pattern highlights persistent technological fragmentation within the Single Market.

A comparison of AI adoption and B2C e-commerce across countries shows a clear positive association. Economies with higher levels of AI adoption tend to report a larger share of firms engaged in online consumer sales. While dispersion across countries remains substantial, the overall relationship suggests that stronger digital capabilities are linked to more developed cross-border e-commerce activity. When a more restrictive indicator of AI intensity - firms using at least two AI technologies is applied - the relationship remains qualitatively similar, although adoption at this level is low across most member states.

Time-series evidence for selected countries illustrates heterogeneous development paths. Digital frontrunners such as Denmark and Finland exhibit faster and more stable growth in B2C e-commerce, while countries with lower AI adoption, including Romania, show slower progress. Poland follows an intermediate trajectory. A comparison with B2B e-commerce indicates a weaker association with AI adoption, suggesting that AI is more closely related to consumer-facing digital trade. Exchange rate developments provide additional context, reflecting the uncertain macroeconomic environment in which these cross-border dynamics unfold.

4. Discussion and policy implications

The findings suggest that the diffusion of artificial intelligence is closely linked to the development of cross-border digital trade within the European Union. Countries with higher levels of AI adoption tend to achieve stronger outcomes in B2C e-commerce, while economies lagging behind in digital capabilities participate less in consumer-oriented online trade. This pattern suggests that AI adoption can act as a reinforcing mechanism, potentially widening existing differences across member states if diffusion remains uneven.

From a policy perspective, the results highlight the risk that uneven technological adoption could reinforce economic fragmentation within the Single Market. While digital frontrunners benefit more strongly from cross-border e-commerce, countries with lower AI uptake may face increasing difficulties in fully participating in digital trade. The weaker association observed for B2B e-commerce further suggests that AI-related advantages are particularly relevant in consumer-facing activities, where data-driven tools, automation and digital platforms play a larger role.

These findings underline the importance of coordinated EU-level digital and innovation policies. Supporting the diffusion of AI technologies, especially among smaller firms and in lagging regions, may help broaden participation in cross-border e-commerce and strengthen the resilience of the Single Market in an environment characterised by heightened uncertainty and external shocks.

5. Conclusion

This report examined the relationship between artificial intelligence adoption and cross-border e-commerce across EU member states in a fragmented and uncertain economic environment. Using external statistics from Eurostat and the European Central Bank, the analysis showed substantial cross-country differences in AI uptake and digital trade outcomes.

The results point to a positive association between AI adoption and B2C e-commerce, with digitally advanced economies achieving stronger participation in cross-border online trade. This relationship remains robust when a more restrictive measure of AI intensity is considered, while the link with B2B e-commerce appears weaker. Differences in development paths over time further highlight persistent digital fragmentation within the European Union.

Overall, the findings suggest that AI diffusion plays an important role in shaping cross-border digital integration. At the same time, uneven adoption risks reinforcing existing disparities across member states, particularly in an environment characterised by heightened uncertainty. Addressing these differences remains a key challenge for policies aimed at strengthening the Single Market in the digital era.

Data sources:

- https://data.ecb.europa.eu/data/datasets/EXR/EXR.D.GBP.EUR.SP00.A
- https://data.ecb.europa.eu/data/datasets/EXR/EXR.D.CNY.EUR.SP00.A
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